



Fig. 4: Visualization of the segmentation output. Top row: Soft segmentation output as red blue plot. The intensity of the plot reflects the confidence. Bottom row hard class labels.

Task: Metric	MaxF1	AP
VGG-pool5	95.80 %	92.19 %
ResNet50	95.89 %	92.10 %
VGG-fc7	95.94 %	92.24 %
ResNet101	96.29 %	92.32 %

TABLE II: Performance of the segmentation decoder.

Task: Metric	moderate	easy	hard
VGG no RIO pool	77.00 %	86.45 %	60.82 %
Faster-RCNN	78.42 %	91.62 %	66.85 %
VGG-pool5	84.76 %	92.18 %	68.23 %
ResNet50	86.63 %	95.55 %	74.61 %
ResNet101	89.79 %	96.13 %	77.65 %

TABLE III: Performance of our detection decoder.

	speed [msec]	speed [fps]
VGG-pool5	42.14 ms	23.73 Hz
ResNet50	39.56 ms	25.27 Hz
VGG-fc7	96.84 ms	10.32 Hz
ResNet101	69.91 ms	14.30 Hz

TABLE IV: Inference speed of our segmentation.

	speed [msec]	speed [fps]	processing
VGG no RIO	35.75 ms	27.96 Hz	2.46 ms
Faster-RCNN	78.42 ms	12.75 Hz	5.3 ms
VGG-pool5	37.31 ms	26.79 Hz	3.61 ms
ResNet50	40.09 ms	24.93 Hz	3.19 ms
ResNet101	65.89 ms	15.17 ms	3.11 ms

TABLE V: Inference speed of our segmentation detection.

V. EXPERIMENTAL RESULTS

In this section we perform our experimental evaluation on the challenging KITTI dataset.

A. Dataset

We evaluate MultiNet on the KITTI Vision Benchmark Suite [14]. The Benchmark contains images showing a variety of street situations captured from a moving platform driving around the city of Karlsruhe. In addition to the raw data, KITTI comes with a number of labels for different tasks relevant to autonomous driving. We use the road benchmark of [12] to evaluate the performance of our semantic segmentation decoder and the object detection benchmark [15] for the detection decoder. We exploit the automatically generated labels of [37], which provide us with road labels generated by combining GPS information with open-street map data.

Detection performance is measured using the average precision score [11]. For evaluation, objects are divided into three categories: easy, moderate and hard to detect. The segmentation performance is measured using the MaxF1 score [12]. In addition, the average precision score is given for reference. Classification performance is evaluated by computing the mean accuracy, precision and recall.

B. Experimental evaluation

The section is structured as follows. We first evaluate the performance of the three decoders individually. To do this we fine-tune the encoder using just one of the three losses

segmentation, detection and classification and compare their performance with a variety of baseline. In the second part we compare joint training of all three decoders with individual inference and show, that the performance of joint training can keep up with the performance of individual inferences. Overall we show, that our approach is competitive with individual inference. This makes our approach very relevant. Joint training has many advantages in robotics application, such as a fast inference time.

a) Segmentation: The segmentation decoder encoder is trained using the four different encoders discussed in Section III-A. The scores, computed on a halt-out validation set is reported in Table I.

To compare our approach against the state-of-the-art we trained a segmentation network with VGG-fc7 encoder on the whole training set and submitted the results to the KITTI road leaderboard. At submission time our approach achieved first place in the benchmark. Recently our approach was overtaken by newer submissions. Some of those are directly based on our work and utilize our code [52]. The results at the time of our benchmark submission are given in Table II.

Qualitative results are shown in Fig. 4 both as red blue plot showing the confidence level at each pixel as well as a hard prediction using a threshold of 0.5.

b) Detection: The detection decoder is trained and evaluated on the data provided by the KITTI object benchmark [15]. We train the detection decoder on a VGG [53] and ResNet [22] decoder and evaluate on a validation set.